

Rishitha Kandi

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Professional Summary

Controls Engineer with 4+ years of hands-on experience in industrial automation, PLC programming, SCADA/HMI development, commissioning, and production-floor troubleshooting in high-volume manufacturing. Experienced with Allen-Bradley ControlLogix/CompactLogix, Siemens and Omron PLCs, Studio 5000, TIA Portal, SIMATIC Manager, Sysmac Studio, CX-Programmer, Ignition SCADA, FactoryTalk View, Cognex machine vision, FANUC PMC, industrial networks, conveyors, drives, motion systems, and automated equipment. Skilled in fault isolation, root-cause analysis, controls modifications, and validation to improve uptime, quality, and production reliability.

Technical Skills

PLC Programming & Automation: Allen-Bradley ControlLogix & CompactLogix | Siemens S7 | Omron PLCs | FANUC PMC | Ladder Logic | Sequencing | Interlocks | Permissives | Timers & Counters

Rockwell / Allen-Bradley: Studio 5000 Logix Designer | FactoryTalk View ME/SE | EtherNet/IP | Distributed I/O | PLC/HMI Troubleshooting | I/O Checkout

Siemens: SIMATIC Manager / STEP 7 | TIA Portal | STARTER | HMI Pro CS | PROFINET | PLC & Drive Diagnostics

Omron: Sysmac Studio | CX-Programmer | Network Configurator | OS32C Configuration Tool | DeviceNet | PLC/HMI Diagnostics

Ignition SCADA: Ignition Gateway | Ignition Designer | Vision Client Launcher | Tags | Alarms | Trends | Historical Data | PLC-SCADA Integration

Drives & Motion: IndraWorks | Bosch Rexroth BODAC | Motion Pro | DriveTop | STARTER | Servo Drives | VFDs | Position Feedback

Vision & Industrial Tools: Cognex In-Sight Explorer | Cognex DataMan | Atlas Copco ToolsTalk | DWG TrueView | Vision Inspection | Error-Proofing

Networks / Electrical / Safety: EtherNet/IP | PROFINET | DeviceNet | Modbus TCP/RTU | Sensors | Motors | Conveyors | Electrical Schematics | LOTO | NFPA 70E | FAT | SAT | Commissioning

Professional Experience

General Motors | Controls Engineer

Moraine, OH | Feb 2026 – Present

- Troubleshoot Omron and Siemens PLC-controlled automated manufacturing equipment by tracing ladder logic, machine sequences, permissives, interlocks, timers, counters, and live I/O during production downtime.
- Diagnose conveyor, motor, actuator, and drive faults by evaluating PLC commands, sensor feedback, position signals, safety permissives, drive status, and HMI alarms.
- Support Cognex machine vision and identification systems, troubleshooting camera triggers, inspection results, NOK conditions, PLC handshakes, DataMan readers, communications, and error-proofing logic.
- Troubleshoot Omron PLC, HMI, DeviceNet, and safety systems using CX-Programmer, Sysmac Studio, Network Configurator, and OS32C Configuration Tool.
- Diagnose Siemens PLC and drive-controlled equipment using SIMATIC Manager, TIA Portal, STARTER, and HMI diagnostics, tracing PLC commands, drive status, position feedback, and field signals.
- Troubleshoot material-handling, conveyor, and positioning systems by analyzing commanded positions, position-complete signals, sensors, actuators, drive feedback, and sequence conditions.
- Perform production-floor troubleshooting of sensors, photoeyes, proximity switches, actuators, motors, drives, and analog/digital I/O to isolate controls, electrical, and mechanical failures.
- Collaborate with Quality, Production, Maintenance, and Manufacturing Engineering on root-cause investigations, controls modifications, equipment recovery, and validation testing.

Rockwell Automation | Controls Engineer

USA | Jan 2025 – Feb 2026

- Developed, modified, and troubleshoot Allen-Bradley ControlLogix and CompactLogix PLC programs in Studio 5000, including ladder logic, sequencing, timers, counters, permissives, interlocks, alarms, and fault-recovery logic.
- Configured and troubleshoot Allen-Bradley EtherNet/IP networks and distributed I/O, diagnosing controller communication, remote I/O, device connectivity, and field-device issues.
- Developed and modified FactoryTalk View ME/SE HMI applications with equipment status, alarms, operator controls, diagnostic displays, fault messages, and PLC tag integration.
- Integrated Allen-Bradley PLCs with VFDs, motors, conveyors, sensors, photoeyes, actuators, and automated material-handling equipment.
- Performed PLC/HMI commissioning, FAT, SAT, I/O checkout, equipment start-up, and controls validation, troubleshooting wiring, PLC logic, field I/O, communications, and sequencing issues.

- Diagnosed recurring Allen-Bradley automation faults using Studio 5000 online monitoring, FactoryTalk diagnostics, electrical schematics, and field-device testing.

University of Dayton | Controls & Automation Labs Teaching Assistant *Dayton, OH | Sep 2023 – Dec 2024*

- Supported PLC/HMI automation labs covering ladder logic, sequencing, alarms, sensors, operator interfaces, analog/digital I/O checkout, motor controls, interlocks, and control-panel troubleshooting.
- Maintained electrical schematics and controls documentation while reinforcing LOTO, NFPA 70E, and OSHA-aligned electrical safety practices.

Wipro | Controls Engineer

Hyderabad, India | Jan 2022 – May 2023

- Developed and maintained Ignition SCADA/HMI applications for industrial automation, creating operator screens, equipment-status displays, navigation, process visualization, and real-time production monitoring.
- Built HMI screens using Ignition Designer and configured PLC tags for real-time equipment status, sensor conditions, process values, and production faults.
- Configured Ignition tags, alarms, alarm history, trends, and historical data logging to support production monitoring and troubleshooting.
- Integrated Ignition with industrial PLCs and field devices, validating PLC-to-SCADA data exchange and troubleshooting communication and tag-quality conditions.
- Developed reusable HMI components and parameterized displays for motors, conveyors, sensors, valves, and production equipment.
- Supported application deployment and troubleshooting through Ignition Gateway, Ignition Designer, and Vision Client Launcher during commissioning, testing, and production support.

Projects

CNC Oil-Hole Drilling Tool Inspection & Quality Improvement

- Investigated an oil-hole drilling quality issue on two CNC machining operations after improperly machined engine components escaped production inspection.
- Reproduced the failure using an empty tool holder and confirmed the machine continued without generating the expected Tool Inspection fault for a missing or broken drill.
- Verified the physical tool-checking mechanism and traced the root cause to CNC program logic where the required tool-inspection M-code had been commented out or bypassed.
- Corrected the M-code logic and validated normal and missing-tool conditions, restoring proper Tool Inspection fault generation.
- Worked with Quality, Production, and Maintenance to verify the correction and strengthen broken-tool detection to prevent improperly drilled components from progressing downstream.

Ignition SCADA – Production Monitoring & Alarm Management

- Developed an Ignition SCADA application for centralized visualization and monitoring of automated manufacturing equipment and production processes.
- Built operator HMI screens in Ignition Designer and configured PLC tags for real-time machine status, sensor conditions, process parameters, and equipment faults.
- Configured alarms, alarm history, trends, and historical data to support troubleshooting and analysis of recurring production conditions.
- Established and troubleshot PLC-to-Ignition communication, validating tag values and resolving communication and tag-quality issues.
- Deployed and tested the application through Ignition Gateway and Vision Client Launcher for system validation, operator access, and production monitoring.

Certifications

Ignition Core Certified | 30-Hour OSHA General Industry Safety and Health | Studio 5000 Advanced Ladder Logic Training | RSLogix/Studio 5000 Ladder Logic Training | Master Allen-Bradley PLC RSLogix/Studio 5000 Essential Guide | Siemens PLC and TIA Portal Essentials | HMI Design and Intermediate PLC Programming | Advanced Programming, Control, and Monitoring Systems | Learn PLC and HMI from Scratch

Education

University of Dayton | Master of Science in Computer Engineering

Dayton, OH | Dec 2025

- Relevant Coursework: Control Systems, Embedded Systems, Contemporary Digital Systems, Digital IC Design, Analog Integrated Circuit Design, Computer Design, Electronics Systems Lab, Digital Signal Processing.